

Zirconium-mediated Selective Synthesis of 1,4-Dialkyl(aryl)-hexa-substituted Benzenes from Two Silyl-substituted Alkynes and One Internal Alkyne

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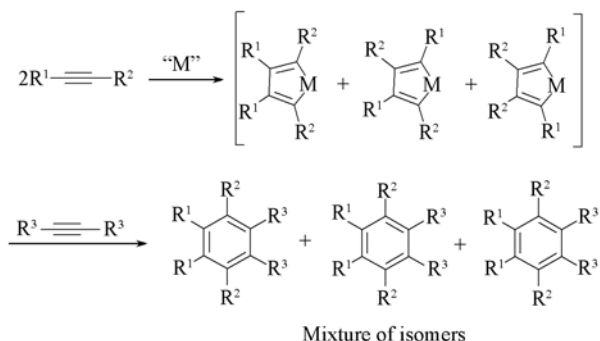
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Abstract Based on the formation of 2,5-bis(trimethylsilyl)-zirconacyclopentadienes from two silylalkynes with Cp_2ZrBu_2 and the Cu-mediated formation of 1,4-disilylbenzene *via* the cycloaddition of zirconacyclopentadienes to disubstituted alkynes, a selective synthesis of 1,4-dialkyl(aryl)-hexa-substituted benzenes was achieved followed by iodination and coupling reaction. The coupling reactions were carried out with either organolithium reagent or organozinc reagent (Negishi coupling), depending on the electrophilic species.

Keywords Zirconium; Hexa-substituted benzene; Iodination; Coupling reaction

1 Introduction

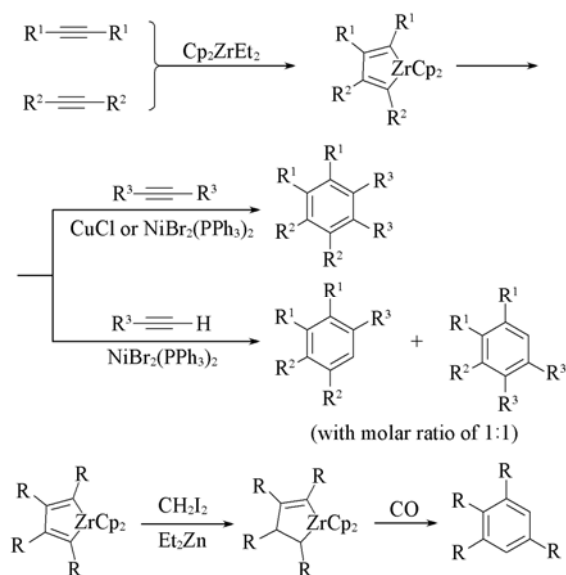
Benzene derivatives are widely utilized organic materials, which become potentially useful in electronic devices^[1–5], biomedicine^[6–9] and chiral ligands^[10–14]. The [2+2+2] cyclotrimerization of three alkynes coupled with transition-metal catalyzing or mediating to give benzene derivatives has been well established in organic synthesis^[15–17]. Among these methods, the syntheses of hexa-substituted benzenes from different alkynes *via* cyclotrimerization mediated by diverse transition metals have been extensively studied for their good compatibility of substrates^[18–22]. However, a major problem remained virtually unsolved is the difficulty in regioselective intermolecular cyclotrimerization of unsymmetrical alkynes with transition-metal to give multisubstituted benzene derivatives. As shown in Scheme 1, when two unsymmetrical alkynes react with transition metals, the corresponding



Scheme 1 Formation of a mixture of benzene derivative isomers mediated by transition metals

metalacycles can be obtained as a mixture of three possible regioisomers^[23–27]. The following cyclization with one internal alkyne affords a mixture of three benzene derivatives.

In our previous works^[28–31], we reported the cyclotrimerization of alkynes mediated by zirconium, as a straightforward route to multi-substituted benzenes (Scheme 2). When CuCl was used, at least one electron-withdrawing group was required for the third alkyne. In other cases, the third alkyne with electron-donating groups (alkyl, trimethylsilyl, alkoxy, etc.) could be introduced by the $\text{NiBr}_2(\text{PPh}_3)_2$ catalyst.



Scheme 2 Synthesis of multi-substituted benzenes mediated by zirconium

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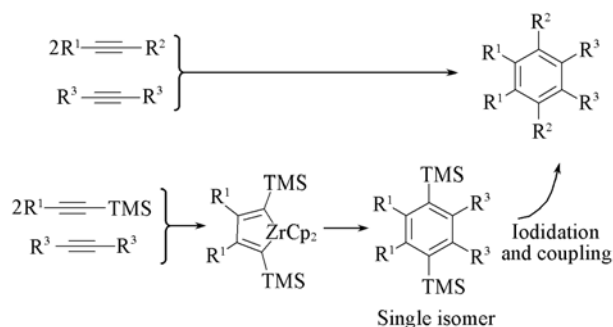
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There have been few reports on the preparation of hexa-substituted benzene derivatives from alkynes with 1,4-alkyl or aryl groups, mainly because the metallocycle can be formed with two unsymmetrical alkynes in a poor regioselectivity. Therefore, one reasonable strategy to overcome this difficulty is to develop a reaction, by which only single isomer easily to be used for further C—C bond formations can be produced. Moreover, it is worth noting that the reaction of $\text{Cp}_2\text{Zr(II)}$ species with two silylalkynes can afford 2,5-disilyl-substituted zirconacyclopentadiene as the single product in a high yield with excellent regioselectivity. So silylalkyne seems to be a suitable substrate due to the characteristic property mentioned above. Based on this, in our previous work^[32], we reported the cycloaddition reaction of 2,5-disilylzirconacyclopentadiene with an internal alkyne to give 1,4-disilyl-hexa-substituted benzene, which could give 1,2,4,5-tetrasubstituted benzenes *via* further desilylation.

Therefore, we reported a method started with the cyclotrimerization of two silyl-substituted alkynes and one internal alkyne to give 1,4-disilyl-hexa-substituted benzene with excellent regioselectivity as a single isomer. After iodination and coupling, the corresponding 1,4-dialkyl(aryl)-hexa-substituted benzene was obtained in a moderate to high yield (Scheme 3). The coupling reactions were studied with organolithium and organozinc reagents for alkyl and aryl electrophiles.



Scheme 3 Synthesis of 1,4-dialkyl(aryl)-hexa-substituted benzenes

2 Experimental

2.1 General Considerations

Unless otherwise noted, all the reactions including air and moisture-sensitive materials were carried out under the protection of nitrogen atmosphere using standard Schlenk techniques. Tetrahydrofuran (THF) was distilled over sodium and benzophenone under the protection of nitrogen atmosphere. 3,6-Bis-trimethylsilyl-4,5-dimethyl-phthalic acid dimethyl ester (**1a**), 3,6-bis-trimethylsilyl-4,5-dipropyl-phthalic acid dimethyl ester (**1b**), 3,6-bis-trimethylsilyl-4,5-dibutyl-phthalic acid dimethyl ester (**1c**), 3,6-bis-trimethylsilyl-4,5-dipentyl-phthalic acid dimethyl ester (**1d**) were given *via* the literature procedures^[32]. All other reagents were commercially available unless otherwise stated. ^1H and ^{13}C NMR spectra were recorded on a JEOL JNM-AL300 spectrometer (FT, 300 MHz for ^1H ; 75 MHz for ^{13}C) at room temperature unless otherwise

noted. Thin layer chromatography (TLC) was performed using Merck Silica Gel 60 F254 plates. Gas chromatography (GC) detection was performed on a Shimadzu GC-14B (FID detector). The NMR characterization was listed in the Electronic Supplementary Material of this paper.

2.2 General Synthetic Procedure for Compounds 2—4

2.2.1 General Synthetic Procedure for 3,6-Diiodo-phthalic Acid Dimethyl Esters (**2a**—**2d**)

To a solution of 3,6-bis-trimethylsilyl-phthalic acid dimethyl esters **1** (1 mmol) in 5 mL of CH_2Cl_2 in a 25 mL Schlenk tube was added the ICl solution (1.0 mol/L CH_2Cl_2 solution, 2.4 mL, 2.4 mmol) dropwise over 10 min at 0 °C. After stirring for 6 h, the absence of compound **1** was confirmed by TLC analysis. The mixture was quenched with a 10% (mass ratio) aqueous $\text{Na}_2\text{S}_2\text{O}_3$ solution, and extracted with CHCl_3 three times. The combined organic layer was dried over MgSO_4 and evaporated to dryness. The residue was purified by a flash column chromatography [silica gel, $V(\text{hexane}):V(\text{ethyl acetate})=10:1$ as eluent] to afford compounds **2**.

3,6-Diiodo-4,5-dimethyl-phthalic acid dimethyl ester (**2a**), a colorless solid, ^1H NMR (300 MHz, CDCl_3), δ : 2.65 (s, 6H), 3.90 (s, 6H); ^{13}C NMR (75 MHz, CDCl_3), δ : 28.7, 53.0, 100.6, 137.7, 142.6, 167.6; HRMS (EI), m/z , calcd. for $\text{C}_{12}\text{H}_{12}\text{I}_2\text{O}_4$: 473.8825; found: 473.8828.

3,6-Diiodo-4,5-dipropyl-phthalic acid dimethyl ester (**2b**), a colorless solid, ^1H NMR (300 MHz, CDCl_3), δ : 1.016 (t, $J=7.0$ Hz, 6H), 1.500 (m, 4H), 2.870 (m, 4H), 3.843 (s, 6H); ^{13}C NMR (75 MHz, CDCl_3), δ : 14.477, 23.029, 42.326, 53.160, 100.715, 138.499, 146.473, 167.965; HRMS (EI), m/z , calcd. for $\text{C}_{12}\text{H}_{12}\text{I}_2\text{O}_4$: 529.9451; found: 529.9454.

3,6-Diiodo-4,5-dibutyl-phthalic acid dimethyl ester (**2c**), a colorless solid, ^1H NMR (300 MHz, CDCl_3), δ : 0.99 (t, $J=7.0$ Hz, 6H), 1.38—1.59 (m, 8H), 2.85—3.04 (m, 4H), 3.89 (s, 6H); ^{13}C NMR (75 MHz, CDCl_3), δ : 13.7, 22.9, 31.3, 39.8, 53.0, 100.4, 138.1, 146.2, 167.6; HRMS (EI), m/z , calcd. for $\text{C}_{18}\text{H}_{24}\text{I}_2\text{O}_4$: 557.9764; found: 557.9768.

3,6-Diiodo-4,5-dipentyl-phthalic acid dimethyl ester (**2d**), a colorless solid, ^1H NMR (300 MHz, CDCl_3), δ : 0.94 (t, $J=7.0$ Hz, 6H), 1.33—1.56 (m, 12H), 2.83—3.02 (m, 4H), 3.89 (s, 6H); ^{13}C NMR (75 MHz, CDCl_3), δ : 14.0, 22.2, 29.0, 32.0, 40.1, 53.0, 100.4, 138.1, 146.2, 167.6; HRMS (EI), m/z , calcd. for $\text{C}_{20}\text{H}_{28}\text{I}_2\text{O}_4$: 586.0077; found: 586.0078.

2.2.2 General Synthetic Procedure for 3-Butyl-4,5-dialkyl-phthalic Acid Dimethyl Esters (**4a** and **4b**)

To a solution of 3,6-diiodo-phthalic acid dimethyl esters **2** (1.0 mmol) in 5 mL of THF in a 25 mL Schlenk tube was added an *n*-BuLi solution (1.65 mol/L *n*-hexane solution, 1.33 mL, 2.2 mmol) dropwise over 10 min at -78 °C and the mixture was stirred for 1 h. Then 1,3-dimethyl-3,4,5,6-tetrahydro-2(1*H*)-pyrimidinone (DMPU, 10 mmol, 1.21 mL) was added. The reaction was kept at about -30 °C for 3 h. The mixture was quenched with 3 mol/L HCl solution, and extracted with ethyl acetate three times. The combined organic layer was washed

with a saturated NaHCO_3 solution and brine, dried over anhydrous MgSO_4 and evaporated to dryness. The residue was purified by a flash column chromatography[silica gel, $V(\text{hexane}):V(\text{ethyl acetate})=10:1$ as eluent] to afford compounds **4**.

3-Butyl-4,5-dipropyl-phthalic acid dimethyl ester(**4a**), a colorless solid, ^1H NMR(300 MHz, CDCl_3), δ : 0.92(m, 9H), 1.43(m, 2H), 1.56(m, 6H), 2.54(m, 6H), 3.85(s, 6H), 7.66(s, 1H); ^{13}C NMR(75 MHz, CDCl_3), δ : 13.0, 13.5, 14.3, 23.5, 24.5, 30.7, 31.4, 33.8, 35.2, 52.4, 125.1, 129.1, 133.8, 138.8, 142.6, 145.0, 166.9, 171.1; HRMS(EI), m/z , calcd. for $\text{C}_{12}\text{H}_{12}\text{O}_4$: 334.2144; found: 334.2144.

3,4,5-Tributyl-phthalic acid dimethyl ester(**4b**), a colorless solid, ^1H NMR(300 MHz, CDCl_3), δ : 0.88—1.03(m, 9H), 1.34—1.60(m, 12H), 2.50—2.72(m, 6H), 3.85(s, 3H), 3.91(s, 3H), 7.66(s, 1H); ^{13}C NMR(CDCl_3), δ : 13.7, 13.8, 13.9, 22.8, 23.28, 23.32, 28.7, 30.5, 32.7, 33.0, 33.5, 33.6, 52.18, 52.24, 124.8, 128.7, 133.4, 138.4, 142.4, 144.7, 166.5, 170.7; HRMS(EI), m/z , calcd. for $\text{C}_{22}\text{H}_{34}\text{O}_4$: 362.2457; found: 362.2456.

2.2.3 General Synthetic Procedure for Hexa-substituted Benzenes(**3c—3h**) with Compounds **4c** and **4d** as Side Products

To a solution of 3,6-diiodo-phthalic acid dimethyl esters **2**(0.4 mmol) in 3 mL of THF in a 25 mL Schlenk tube was added $t\text{-BuLi}$ (1.65 mol/L $n\text{-hexane}$ solution, 1.07 mL, 1.76 mmol) dropwise over 10 min at -78°C . The mixture was stirred for 1 h. Then DMPU(4 mmol, 0.48 mL) and electrophile iodoalkane(1.6 mmol) were added to the solution. The mixture was kept at 0°C for 3 h and stirred at room temperature for another 3 h. Then the mixture was quenched with 3 mol/L HCl and extracted with ethyl acetate three times. The combined organic phase was washed with DI water, a saturated NaHCO_3 solution, and brine, dried over anhydrous MgSO_4 and evaporated to dryness. The residue was purified by a flash chromatography[silica gel, $V(\text{hexane}):V(\text{ethyl acetate})=10:1$ as eluent] to afford compounds **3**.

3,6-Dipentyl-4,5-dimethyl-phthalic acid dimethyl ester(**3c**), a colorless solid, ^1H NMR(300 MHz, CDCl_3), δ : 0.90(t, $J=7.2$ Hz, 6H), 1.22—1.44(m, 8H), 1.44—1.63(m, 4H), 2.25(s, 6H), 2.55—2.72(m, 4H), 3.83(s, 6H); ^{13}C NMR(75 MHz, CDCl_3), δ : 14.0, 16.3, 22.4, 30.2, 31.3, 32.2, 52.1, 129.8, 136.7, 138.1, 139.8; HRMS(EI), m/z , calcd. for $\text{C}_{22}\text{H}_{34}\text{O}_4$: 362.2457; found: 362.2453.

3-Pentyl-4,5-dimethyl-phthalic acid dimethyl ester(**4c**), a colorless solid, ^1H NMR(300 MHz, CDCl_3 , Me_4Si), δ : 0.90(t, $J=7.0$ Hz, 3H), 1.28—1.56(m, 6H), 2.26(s, 3H), 2.32(s, 3H), 2.52—2.61(m, 2H), 3.86(s, 3H), 3.92(s, 3H), 7.66(s, 1H); ^{13}C NMR(75 MHz, CDCl_3), δ : 14.0, 15.6, 20.8, 22.3, 30.0, 31.4, 32.2, 52.2, 52.3, 129.0, 133.5, 138.1, 138.4, 140.8, 166.4, 170.6, 182.7; HRMS(EI), m/z , calcd. for $\text{C}_{17}\text{H}_{24}\text{O}_4$: 292.1675; found: 292.1675.

3,6-Dipentyl-4,5-dibutyl-phthalic acid dimethyl ester(**3d**), a colorless solid, ^1H NMR(300 MHz, CDCl_3 , Me_4Si), δ : 0.90(t, $J=6.8$ Hz, 6H), 0.98(t, $J=6.3$ Hz, 6H), 1.25—1.60(m, 20H), 2.53—2.68(m, 8H), 3.83(s, 6H); ^{13}C NMR(75 MHz, CDCl_3), δ : 13.8, 14.0, 22.3, 23.4, 29.1, 30.5, 31.4, 32.4, 33.4, 52.1, 130.3,

137.0, 142.1, 170.0; HRMS(EI), m/z , calcd. for $\text{C}_{28}\text{H}_{46}\text{O}_4$: 446.3396; found: 446.3397.

3-Butyl-4,5-dipentyl-phthalic acid dimethyl ester(**4d**), a colorless solid, ^1H NMR(300 MHz, CDCl_3 , Me_4Si), δ : 0.81—1.05(m, 9H), 1.30—1.68(m, 14H), 2.51—2.68(m, 6H), 3.85(s, 3H), 3.92(s, 3H), 7.65(s, 1H); ^{13}C NMR(75 MHz, CDCl_3), δ : 13.7, 13.92, 13.97, 22.3, 22.8, 23.3, 28.8, 30.8, 31.2, 32.5, 32.7, 33.0, 33.5, 52.2, 53.3, 124.8, 128.9, 133.5, 138.5, 142.3, 145.0, 166.8, 170.9; HRMS(EI), m/z , calcd. for $\text{C}_{23}\text{H}_{36}\text{O}_4$: 376.2614; found: 376.2619.

3,6-Dimethyl-4,5-dibutyl-phthalic acid dimethyl ester(**3e**), a colorless solid, ^1H NMR(300 MHz, CDCl_3), δ : 0.96(t, $J=7.0$ Hz, 6H), 1.35—1.52(m, 8H), 2.29(s, 6H), 2.52—2.71(m, 4H), 3.83(s, 6H); ^{13}C NMR(75 MHz, CDCl_3), δ : 13.7, 16.5, 23.2, 29.5, 32.0, 52.1, 130.3, 131.7, 142.2, 169.6; HRMS(EI), m/z , calcd. for $\text{C}_{20}\text{H}_{30}\text{O}_4$: 334.2144; found: 334.2144.

3,6-Dimethyl-4,5-dipentyl-phthalic acid dimethyl ester(**3f**), a colorless solid, ^1H NMR(300 MHz, CDCl_3), δ : 0.92(t, $J=7.0$ Hz, 6H), 1.29—1.57(m, 12H), 2.29(s, 6H), 2.53—2.75(m, 4H), 3.84(s, 6H); ^{13}C NMR(75 MHz, CDCl_3), δ : 14.0, 16.6, 22.4, 29.6, 29.9, 32.4, 52.2, 130.3, 131.8, 142.4, 169.7; HRMS(EI), m/z , calcd. for $\text{C}_{22}\text{H}_{34}\text{O}_4$: 362.2457; found: 362.2459.

3,6-Dipropyl-4,5-dimethyl-phthalic acid dimethyl ester(**3g**), a colorless solid, ^1H NMR(300 MHz, CDCl_3), δ : 0.97(t, $J=7.3$ Hz, 6H), 1.47—1.62(m, 4H), 2.25(s, 6H), 2.55—2.71(m, 4H), 3.83(s, 6H); ^{13}C NMR(75 MHz, CDCl_3), δ : 14.4, 16.3, 23.8, 33.4, 52.1, 129.9, 136.6, 138.1, 169.8; HRMS(EI), m/z , calcd. for $\text{C}_{18}\text{H}_{26}\text{O}_4$: 306.1831; found: 306.1830.

3,6-Dihexyl-4,5-dibutyl-phthalic acid dimethyl ester(**3h**), a colorless solid, ^1H NMR(300 MHz, CDCl_3), δ : 0.89(t, $J=6.7$ Hz, 6H), 0.98(t, $J=6.7$ Hz, 6H), 1.14—1.54(m, 24H), 2.52—2.67(m, 8H), 3.82(s, 6H); ^{13}C NMR(75 MHz, CDCl_3), δ : 13.9, 14.2, 22.7, 23.6, 29.3, 30.1, 30.7, 31.6, 31.9, 33.5, 52.3, 130.4, 137.2, 142.3, 170.0; HRMS(EI), m/z , calcd. for $\text{C}_{30}\text{H}_{50}\text{O}_4$: 474.3709; found: 474.3711.

2.2.4 General Synthetic Procedure for 3,6-Bis-thienyl-4,5-dimethyl-phthalic Acid Dimethyl Ester(**3i**)

A solution of ZnCl_2 (0.272 g, 2 mmol) was added to a 25 mL Schlenk tube I and heated to melt by a blower *in vacuo* to remove the moisture. After cooling to room temperature, nitrogen was introduced into the flask and THF(3 mL) was added to the Schlenk tube I. Meanwhile, to a solution of thiophene(0.135 mL, 1.7 mmol) in 3 mL of THF in a 25 mL Schlenk tube II was added $n\text{-BuLi}$ (1.65 mol/L $n\text{-hexane}$ solution, 0.97 mL, 1.6 mmol) dropwise over 10 min at -78°C , and the mixture was stirred for 0.5 h. The solution was warmed to 0°C for 1 h. Then the solution of ZnCl_2 in THF was transferred from the Schlenk tube I to the Schlenk tube II at -78°C and stirred for 0.5 h. The solution was warmed to 0°C for 1 h. After that, the solution from the Schlenk tube II was transferred to the Schlenk tube III with 3,6-diiodo-4,5-dimethyl-phthalic acid dimethyl ester(**2a**, 0.190 g, 0.4 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.023 g, 5%) in it. After stirred for 2 h at 50°C , the mixture was quenched with 3 mol/L HCl and extracted with ethyl acetate three times. The combined organic phase was washed with DI water, a saturated NaHCO_3 solution, and brine respectively,

dried over anhydrous MgSO_4 and evaporated to dryness. The residue was purified by a flash chromatography[silica gel, $V(\text{hexane}):V(\text{ethyl acetate})=5:1$ as eluent] to afford compound **3i**.

3,6-Bis-thienyl-4,5-dimethyl-phthalic acid dimethyl ester (**3i**), a purple solid, ^1H NMR(300 MHz, CDCl_3), δ : 2.20(s, 6H), 3.54(s, 6H), 6.83—7.00(m, 2H), 7.00—7.14(m, 2H), 7.33—7.46(m, 2H); ^{13}C NMR(75 MHz, CDCl_3), δ : 18.1, 52.1, 126.3, 126.6, 127.8, 131.5, 132.6, 139.0, 140.0, 168.0; HRMS(EI), m/z , calcd. for $\text{C}_{20}\text{H}_{18}\text{O}_4\text{S}_2$: 386.0647; found: 386.0649.

2.2.5 General Synthetic Procedure for 3,6-Bis-*p*-methylphenyl-4,5-dibutyl-phthalic Acid Dimethyl Ester(**3j**)

A solution of ZnCl_2 (0.272 g, 2 mmol) was added to a 25 mL Schlenk tube I and heated to melt by a blower *in vacuo* to remove the moisture. After cooling to room temperature, nitrogen was introduced into the flask and an anhydrous THF/NMP solution(1:1, volume ratio, 6 mL) was added to the Schlenk tube I. Meanwhile, to a solution of 4-bromotoluene (0.29 g, 1.7 mmol) in 6 mL of THF/NMP solution in a 25 mL Schlenk tube II was added *n*-BuLi(1.65 mol/L *n*-hexane solution, 0.97 mL, 1.6 mmol) dropwise over 10 min at 0°C , and the mixture was stirred for 0.5 h. The solution was warmed to room temperature for 4 h. Then the solution of ZnCl_2 in THF/NMP in the Schlenk tube I was transferred to the Schlenk tube II at -78°C and stirred for 0.5 h. The solution was warmed to 0°C for 1 h and to room temperature for another 1 h. After that, the solution in the Schlenk tube II was transferred to the Schlenk tube III with 3,6-diiodo-4,5-dibutyl-phthalic acid dimethyl ester(**2c**, 0.223 g, 0.4 mmol), $\text{Pd}(\text{PPh}_3)_4$ (0.023 g, 5%) in it and heated to reflux overnight. Then the mixture was quenched with 3 mol/L HCl and extracted with ethyl acetate three times. The combined organic phase was washed with DI water, a saturated NaHCO_3 solution, and brine respectively, dried over anhydrous MgSO_4 and evaporated to dryness. The residue was purified by a flash chromatography[silica gel, $V(\text{hexane}):V(\text{ethyl acetate})=5:1$ as eluent] to afford compound **3j**.

3,6-Bis-*p*-methylphenyl-4,5-dibutyl-phthalic acid dimethyl ester(**3j**), a colorless solid, ^1H NMR(300 MHz, CDCl_3), δ : 0.70(t, $J=7.2$ Hz, 6H), 1.06—1.19(m, 4H), 1.21—1.34(m, 4H), 2.39(s, 6H), 2.45—2.57(m, 4H), 3.42(s, 6H), 7.05—7.24(m, 8H); ^{13}C NMR(75 MHz, CDCl_3), δ : 13.4, 21.2, 22.9, 29.8, 33.0, 51.9, 128.3, 129.3, 130.3, 136.0, 136.7, 139.7, 142.4, 168.8; HRMS(EI), m/z , calcd. for $\text{C}_{32}\text{H}_{38}\text{O}_4$: 486.2770; found: 486.2770.

2.2.6 General Synthetic Procedure for 3,6-Diphenyl-4,5-dibutyl-phthalic Acid Dimethyl Ester(**3k**)

A solution of ZnCl_2 (0.272 g, 2 mmol) was added to a 25 mL Schlenk tube I and heated to melt by a blower *in vacuo* to remove the moisture. After cooling to room temperature, nitrogen was introduced into the flask and an anhydrous THF/NMP solution(1:1, volume ratio, 6 mL) was added to the Schlenk tube I. At -78°C , phenyllithium(0.98 mol/L dibutyl ether solution, 1.63 mL, 1.6 mmol) was added to the solution dropwise over 10 min and stirred for 0.5 h. Then the mixture was warmed to 0°C for 1 h and to room temperature for

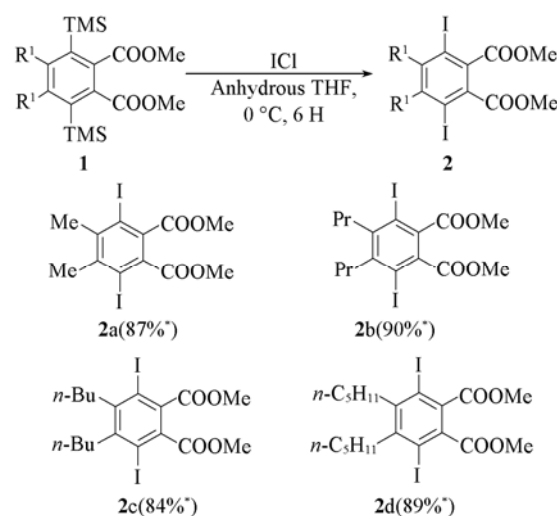
another 1 h. After that, the solution in the Schlenk tube I was transferred to the Schlenk tube II with 3,6-diiodo-4,5-dibutyl-phthalic acid dimethyl ester(**2c**, 0.223 g, 0.4 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.023 g, 5%) in it and heated to reflux overnight. Then the mixture was quenched with 3 mol/L HCl and extracted with ethyl acetate three times. The combined organic phase was washed with DI water, a saturated NaHCO_3 solution, and brine respectively, dried over anhydrous MgSO_4 and evaporated to dryness. The residue was purified by a flash chromatography[silica gel, $V(\text{hexane}):V(\text{ethyl acetate})=5:1$ as eluent] to afford compound **3k**.

3,6-Diphenyl-4,5-dibutyl-phthalic acid dimethyl ester(**3k**), a colorless solid, ^1H NMR(300 MHz, CDCl_3 , Me_4Si), δ : 0.68(t, $J=7.2$ Hz, 6H), 1.03—1.19(m, 4H), 1.21—1.33(m, 4H), 2.42—2.57(m, 4H), 3.38(s, 6H), 7.21—7.43(m, 10H); ^{13}C NMR(75 MHz, CDCl_3), δ : 13.4, 22.9, 29.8, 32.9, 51.9, 127.3, 127.6, 129.4, 130.3, 139.0, 139.8, 142.3, 168.7; HRMS(EI), m/z , calcd. for $\text{C}_{30}\text{H}_{34}\text{O}_4$: 458.2457; found: 458.2458.

3 Results and Discussion

In the cyclotrimerization reaction, substituent groups of the third alkyne could be electron-withdrawing or electron-donating according to the results of our former researches^[32,33]. Based on this, only dimethyl acetylenedicarboxylate(DMAD) was used as a model substrate for the third alkyne in this work. The synthesis of 3,6-disilyl phthalic acid dimethyl esters(**1**) was given in our previous work^[32].

Firstly, it has been reported that silylbenzene served as a starting material in the preparation of a variety of derivatives *via* the reaction with the appropriate electrophile^[34,35]. So we selected iodine monochloride as a source of electrophilic iodine to synthesize certain aromatic iodides or cleave C—Si bonds. As was expected, the iodination of 3,6-disilyl phthalic acid dimethyl ester was efficiently promoted by ICl under mild conditions. 3,6-Diiodo-phthalic acid dimethyl esters(**2**) were carried out in excellent yields(Scheme 4). However, to compare the effect of ICl with those of other electrophiles, no desire

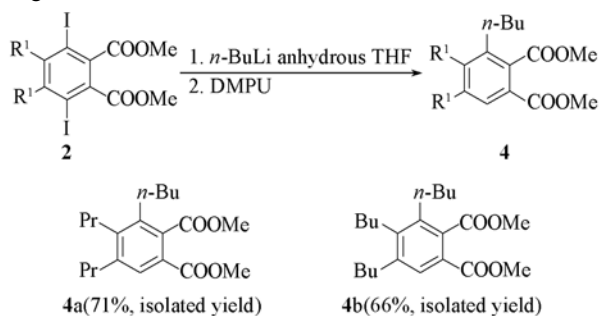


Scheme 4 Iodination of 3,6-disilyl phthalic acid dimethyl ester with iodine monochloride

* Isolated yields.

product was isolated from the reaction mixture when I_2 and Br_2 were used. This was probably due to the difference in the electronegativity between iodine and chlorine. ICl was highly polar and behaved as a better source of I^+ .

Secondly, one important type of coupling reaction is that an organometallic compound RM reacts with an organic halide of type $R'X$ with the formation of a new carbon-carbon bond in the product $R-R'$. In that case, we employed organolithium compound to promote the coupling reaction of 1,4-diiodobenzenes with alkyl or aryl halide. It was the first report using $n-BuLi$ as the lithium reagent. However, no objective product was separated after the reaction was carried out under the following condition of 1 mmol of compound **2** in anhydrous THF, 2.2 mmol of $BuLi$, $-78\text{ }^\circ\text{C}$, 1 h, then 4 mmol of R_2I , $0\text{ }^\circ\text{C}$, 3 h, r.t., 3 h. When R_2I was not added in the control group, and the reaction was quenched at r.t., TLC showed the mixture was so messy. To our surprise, when quenching the reactions at $-30\text{ }^\circ\text{C}$ 3 h later, penta-substituted benzene **4** was obtained with a high yield of about 75% monitored by GC (Scheme 5). This may be resulted from the presence of $n-BuI$ in the reaction mixture, which can react with the organolithium reagent formed.

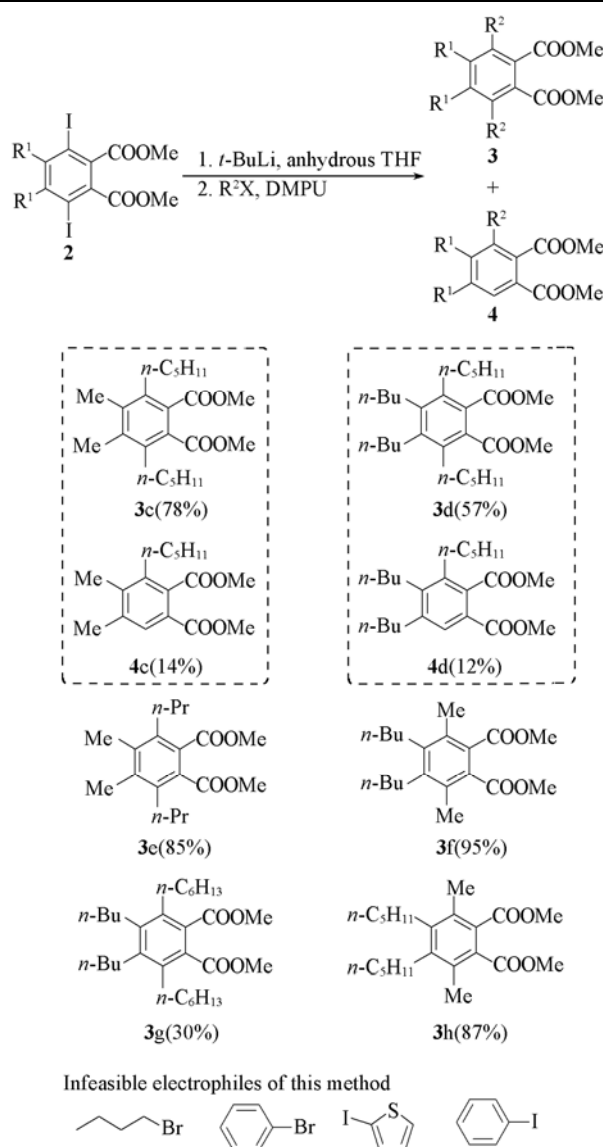


Scheme 5 Coupling reaction of 1,4-diiodobenzenes with $n-BuLi$

Conditions: 1 mmol of 3,6-diiodo-phthalic acid dimethyl esters **2** in anhydrous THF, 2.2 mmol of $n-BuLi$, $-78\text{ }^\circ\text{C}$, 1 h, 10 mmol of DMPU, $-30\text{ }^\circ\text{C}$, 3 h. DMPU was used as a polar, aprotic solvent to promote the coupling reaction. All the yields were isolated yields.

For the case that the formation of hexa-substituted benzene was blocked by the side reaction, $t-BuLi$ rather than $n-BuLi$ was used since the formed $t-BuI$ was immediately destroyed by $t-BuLi$ in a dehydrohalogenation reaction. A series of 1,4-dialkyl-hexa-substituted benzenes was obtained with moderate to excellent yields (Scheme 6), which depended on the substituents of substrates and electrophiles. In rare cases, compounds **4c** and **4d** with a yield of about 10% could be separated as side products due to incomplete reactions. According to the reaction results, if both R^1 and R^2 were long-chain alkyl groups, there will be a significant decrease in the yield of compound **3g** caused by steric effects of *ortho*-position. In addition, R_2X (R =alkyl, aryl; X =Br, I) was examined for feasible electrophiles and only alkyl iodides could promote the reaction smoothly. This illustrated that compared with alkyl iodide, alkyl bromide as an electrophile seemed not to be strong enough to carry out this coupling reaction. In regard to aryl electrophiles, we need to find another method to proceed.

Negishi and coworkers^[36–48] reported the use of palladium

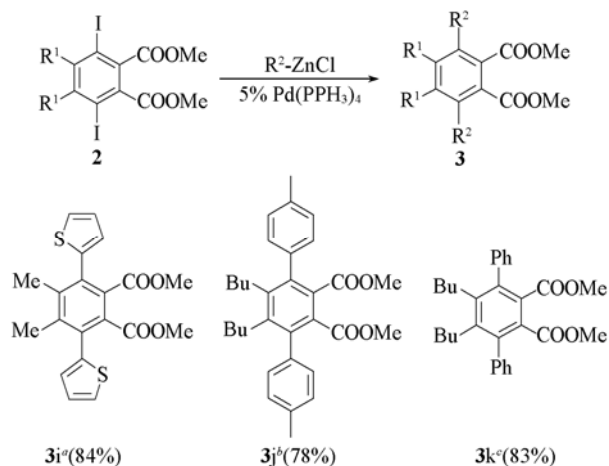


Scheme 6 Coupling reaction of 1,4-diiodobenzenes with alkyl halides by $t-BuLi$

Conditions: 0.4 mmol of 3,6-diiodo-phthalic acid dimethyl esters **2**, 1.76 mmol of $t-BuLi$, anhydrous THF, $-78\text{ }^\circ\text{C}$, 1 h. Then 4 mmol of DMPU, 1.6 mmol of R_2X , $0\text{ }^\circ\text{C}$, 3 h, r.t., 3 h; All the yields were isolated yields.

complexes in organozinc coupling reactions and carried out method studies. Allowing for the coupling of sp_3 , sp_2 , and sp carbons makes the Negishi reaction somewhat unique among the palladium-catalyzed coupling reactions. Therefore, the synthesis of 1,4-diaryl hexa-substituted benzene *via* Negishi coupling reaction was studied (Scheme 7). The active organozinc compound was obtained from the reaction of aryl lithium reagents. For example, with the presence of catalyst $Pd(PPh_3)_4$ (5%), the reaction of compound **2a** with thienyl zinc chloride proceeded smoothly to afford the corresponding 4,5-dimethyl-3,6-bis-thienyl-phthalic acid dimethyl ester (**3i**) as a single product in a noticeable NMR analysis yield of 95%. Regarding the reaction of compound **2c** with phenyl zinc chloride under the condition of pure anhydrous THF, $50\text{ }^\circ\text{C}$, 2 h, no product was obtained due to the limit of temperature. However, though the mixture was heated to reflux for 12 h, the result was

unsatisfied. Thus, mixed solvents were applied to carry out this reaction. When the mixture of THF/*N*-methyl pyrrolidone (NMP)(1:1, volume ratio) was examined and heated to reflux overnight, it was satisfied that compounds **3j** and **3k** were isolated with high yields after column chromatography. The proposed catalytic cycle is illustrated in Fig.1.



Scheme 7 Synthesis of 1,4-diaryl hexa-substituted benzene by Negishi coupling reaction

a. Reaction conditions: 1.7 mmol of thiophen, 1.6 mmol of *n*-BuLi, anhydrous THF, $-78\text{ }^{\circ}\text{C}$, 0.5 h, $0\text{ }^{\circ}\text{C}$, 1 h. Then, 2 mmol of ZnCl_2 , $-78\text{ }^{\circ}\text{C}$, 0.5 h, $0\text{ }^{\circ}\text{C}$, 1 h. Then, 0.4 mmol of 3,6-diiodo-phthalic acid dimethyl ester **2a**, 5% (molar ratio) $\text{Pd}(\text{PPh}_3)_4$, $50\text{ }^{\circ}\text{C}$, 2 h; b. reaction conditions: 1.7 mmol of 4-bromotoluene, 1.6 mmol of *n*-BuLi, anhydrous THF:NMP(1:1, volume ratio), $-78\text{ }^{\circ}\text{C}$, 0.5 h, $0\text{ }^{\circ}\text{C}$, 0.5 h, r.t., 4 h. Then, 2 mmol of ZnCl_2 , $-78\text{ }^{\circ}\text{C}$, 0.5 h, $0\text{ }^{\circ}\text{C}$, 1 h, r.t., 1 h. Then, 0.4 mmol of 3,6-diiodo-phthalic acid dimethyl ester **2c**, 5% (molar ratio) $\text{Pd}(\text{PPh}_3)_4$, reflux, overnight; c. reaction conditions: 2 mmol of ZnCl_2 , 1.6 mmol of PhLi, anhydrous THF: NMP(1:1, volume ratio), $-78\text{ }^{\circ}\text{C}$, 0.5 h, $0\text{ }^{\circ}\text{C}$, 1 h, r. t., 1 h. Then, 0.4 mmol of 3,6-diiodo-phthalic acid dimethyl ester **2c**, 5% (molar ratio) $\text{Pd}(\text{PPh}_3)_4$, reflux, overnight. All these yields were isolated yields.

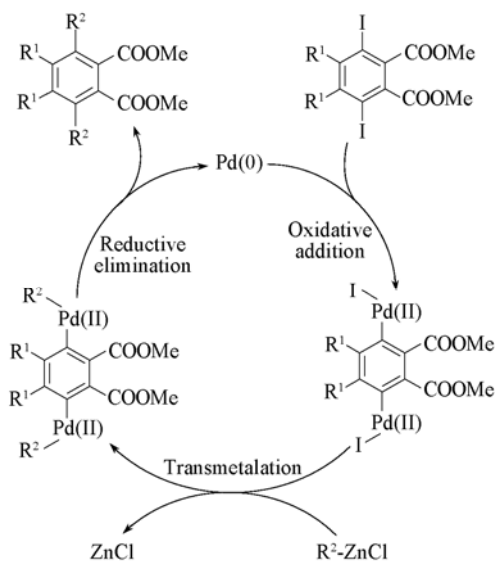


Fig.1 Proposed catalytic cycle for Negishi coupling of 3,6-diiodo-phthalic acid dimethyl esters with organozinc reagents

4 Conclusions

In summary, a high regioselective synthesis method for

1,4-dialkyl(aryl)-hexa-substituted benzenes was developed with the cyclotrimerization of two silyl-substituted alkynes and one internal alkyne mediated by zirconium, followed by iodination and two types of coupling reactions for both alkyl and aryl halides. This was a novel strategy to prepare quite useful benzene derivatives with various kinds of substituents. Ester groups which have strong activity for further reactions could be applied for the synthesis of complicated π -conjugated molecules. These benzenes with unique ability can be widely used in various fields, such as optoelectronic materials and pharmaceutical synthesis. Moreover, we are working on this basis to the synthesis of benzene derivatives with 1,4-alkynyl groups, which may have potential to be used as molecular wires.

Electronic Supplementary Material

Supplementary material is available in the online version of this article at <http://dx.doi.org/10.1007/s40242-016-6071-0>.

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